LLMs Can't Keep Up with Language That's on Fleek

How Training Data Age Stunts Model Vocabulary

Anonymous ACL submission

Abstract

The accuracy of large language models is an integral aspect of their performance. Accuracy is commonly thought of in terms of grammar and informational correctness, but also importantly encompasses the correct usage of vocabulary by models. We argue that this "correct usage" encompasses not just avoiding the use of definitionally incorrect words and phrases, but avoiding the use of irrelevant and dated vocabulary. Large language models inherit not only the knowledge but the temporal vocabulary characteristics of their training data. Because web-scale corpora carry no uniform date weighting, a model trained on a decade of internet text may treat decade-old slang as equivalent to current usage. We ask whether LLMs track the temporal lifecycle of informal vocabulary or whether their slang usage is stunted relative to the pace of real-world linguistic change. Combining a frequency analysis of slang terms in FineWeb (Penedo et al., 2024)—a filtered derivative of Common Crawl—with three complementary analyses of four LLMs, we find that (i) model slang usage is anchored to older, well-attested terms irrespective of recency. The tested models use slang with average peak usage over ten years ago; (ii) even with explicit instruction models display inaccurate awareness of peak slang usage; and (iii) models' parametric knowledge of when a term was popular is systematically dated much earlier than its peak popularity based on the FineWeb corpus. We argue these effects compound as human-authored text is increasingly displaced by model output, and discuss implications for future training-data curation, as well as potential societal impacts from common usage of models that tend to stagnate language.

1 Introduction

1.1 Motivation: Training Data Growth and Its Limits

Large language models have scaled rapidly on ever-growing internet text corpora, and data scale remains a primary lever for capability: compute-optimal training scales tokens in proportion to parameters (Hoffmann et al., 2022), (Zhao et al., 2023). Historically internet traffic and the volume of available training data has continued to grow year over year, making recency a secondary concern (Villalobos et al., 2024; Zhao et al., 2023; Reinsel et al., 2018). Two trends are shifting this picture. First, LLM-generated text constitutes a growing share of the web, threatening to contaminate future training pools with model output rather than authentic human language; Shumailov et al. (2023) show that training on model-generated content induces irreversible *model collapse*, in which the tails of the original distribution vanish. Second, if the supply of genuinely human-authored data decelerates, the median age of training data will rise until eventually the bulk of available text may be years or decades old. It as been demonstrated that models lean on older data (Zhao et al., 2024; Zhang and Choi, 2023; Luu et al., 2022; Cheng et al., 2024), we show that this has effects not only on a model's knowledge of events and information, but on its **view of language itself**. This fundamental flaw in model design leads models to inherently anchor their vocabulary to an earlier linguistic era and fail to track how language actually evolves.

1.2 The Temporal Language Problem

Language changes continuously, with slang and informal vocabulary changing at the fastest rate, often rising and falling within months to a few years. Because internet training corpora carry no uniform date weighting, a model trained on a decade of web text may treat decade-old slang as interchangeable with contemporary usage, due to inherent lack of awareness of the structure of their own training data. The problem is most acute for slang of the ∼2010–2020 era: it is abundant in training data yet not clearly dated, so its representation is governed by total corpus frequency rather than current

relevance (Section 4.1). This is compounded by misalignment in the underlying corpora: Cheng et al. (2024) show that effective training cutoffs often differ substantially from reported ones, and that Common Crawl dumps exhibit non-trivial temporal leakage, with old data persisting in newer dumps and being de facto upsampled.

In this paper we find that **LLM slang usage is temporally flat**: models do not differentiate by era when producing informal language, leaving their freeform output sounding dated due to higher representation of older slang in training corpora. We theorize that the "temporal misalignments" shown in (Cheng et al., 2024) also effect the language used by the model itself. We expect this effect to worsen if the bulk of high-quality, human-authored pretraining data continues to age without intervention.

1.3 Linguistic Framing

To demonstrate a temporal linguistic effect we require a set of words whose popularity has measurably risen and fallen within the span of LLM training data. We focus on *slang*, which we situate against two adjacent classes of words or phrases defined by lexical change:

1. **Neologisms:** words that rise in use and plateau as they enter the standard lexicon (Metcalf, 2002). Our hypothesis predicts that models should also lag in adopting neologisms, but their slower, monotonic trajectories are harder to resolve on the time scale of available data.

2. **"Meme words" / ephemeral slang:** words that spike quickly and fade within months to a few years. Due to the current max age (3 decades) of most colloquial text training data (internet text), this time range is ideal for our study. Slang use is both creative and ephemeral (Eble, 1989; Wu and Sun, 2025), for our study we select words for their ephemerality. By selecting words that peaked in 2010–2020 and declined before our models' reported cutoffs, we can test whether reduced recent representation in training data translates into reduced usage by models.

3. **"Vogue words":** standard-English words that undergo a transient usage spike (e.g. *awesome*, *iconic*, *epic*) coined in (Fowler, 1926); temporally they behave much like neologisms. This category is also examined in this study, but should be differentiated from words that increase in usage frequency due to a new definition, which we attempt to separate. That separation is important: lexical change concentrates in new senses attaching to existing forms rather than in newly coined forms (Hamilton et al., 2016), so a frequency spike is only evidence of *slang* adoption once the slang sense has been isolated from the standard one (Section 4.1.1).

In linguistics, slang is defined by register rather than novelty, but the definition is famously contested (Dumas and Lighter, 1978; Kipfer and Chapman, 2010). The property we rely on to define slang here is *ephemerality*: throughout this paper, "slang" denotes a word with a demonstrably ephemeral usage lifecycle in the sense of Eble (1989). Additionally, in order to demonstrate the effects of models using aging slang, we cite examples of slang tied to in-group identity (Eble, 1989; Earl, 1972), in particular in-group to a specific generation (we often contrast "Millenial" and "Gen-Z" slang usage). The authors would also like to acknowledge that a substantial portion of American internet slang is appropriated African American Vernacular English or language appropriated from the Queer community; we return to this in our discussion of limitations (Section 7).

1.4 Contributions

This paper makes the following contributions:

1. A sense-disambiguated, time-resolved frequency analysis of nineteen slang terms in FineWeb, establishing a ground truth for slang prevalence in LLM training data and a proxy for the popularity of the terms at given time slices (Experiment 1).

2. A direct year-association experiment showing that models' knowledge of slang era is systematically shifted into the past, dissociating what models *know* about slang age from how they *use* it (Experiment 2).

3. A semantic-slot experiment showing that models reliably produce older, well-attested slang but fail on the most recent terms, and that conditioning the prompt on an explicit year does not recover era-appropriate vocabulary (Experiment 3).

4. An analysis of slang words generated by model persona simulation, mapping them back to corpus age to demonstrate a model "vintage" based on average age of slang used in a naturalistic setting (Experiment 4).

2 Background & Related Work

2.1 Training Data Sources and Temporal Coverage

Common Crawl is a major component of most large-scale training corpora and a rich source of informal language: it includes web text from platforms such as Reddit[1], whose posts and comments capture a large variety of informal language and internet slang, in contrast to sources like BookCorpus that are dominated by edited formal prose. Owing to Common Crawl's size and noise, a number of cleaned derivatives are distributed on HuggingFace using aggressive deduplication and quality filtering. We examine one popular derivative, FineWeb (Penedo et al., 2024), which provides high-quality internet text using a set of novel and established filtering methods (See https://huggingface.co/datasets/HuggingFaceFW/fineweb#data-processing-steps).

Common Crawl dumps date back to 2013 and have been collected every few months since (Common Crawl Foundation, 2007). FineWeb retains a `dump` field linking each document to its originating crawl. The repository releases pre-sampled subsets at several scales, and our analysis uses the 10-billion-token (10BT) sample as a representative slice. This sample was generated in 2024 and so spans dumps up to `CC-MAIN-2024-10` (the tenth week of 2024). The `dump` field lets us bin documents by date, typically to the granularity of a few months, and measure how the frequency of each target slang term varies over time. This is used as a corpus-level baseline for slang prevalence on the web, and a means of verifying that each target word is genuinely ephemeral (frequent in one date range and scarce in a later one). Additionally, this analysis gives us insights into which words have recently become popular and which have recently fallen out of favor. Common Crawl itself is also the source of many open questions around the representativeness of data used to traing LLMs (Baack, 2024), (Luccioni and Viviano, 2021).

[1]Reddit closed it's API to crawlers in 2024, which is the end date of the FineWeb CommonCrawl basis in this study.

2.2 Temporal Misalignment

This paper is a direct extension of work on *temporal misalignment*: the mismatch between the period a model's training data was drawn from and the period in which the model is deployed.

Measuring temporal misalignment. Luu et al. (2022) formalize the problem by training and evaluating models on time-sliced splits of several domains, showing that performance degrades as the train–test time gap widens, that the rate of degradation is strongly task-dependent, and that neither model scale nor continued pretraining eliminates it. Agarwal and Nenkova (2022) reach a compatible conclusion for pretrained encoders on standard NLP tasks, and further localize the damage: deterioration concentrates on items whose *surface forms* are new to the model, for example vocabulary, rather than on syntax or task structure, which can be partly repaired by continued pretraining on text from the evaluation period.

Where the misalignment originates. Cheng et al. (2024) supply the mechanism on the data side, showing that a model's *effective* cutoff frequently precedes its reported one, because Common Crawl dumps leak old documents forward and imperfect deduplication effectively upsamples them. What governs how strongly a period is represented is the volume of text surviving from each slice of time, a central mechanism explored in this paper.

From facts to language. We apply this framework one level down: to the lexicon a model speaks in rather than the propositions it asserts. This is a change of signal rather than of subject, and each construct above has a lexical counterpart that we measure.

1. **A ground truth indexed by time.** Where Luu et al. (2022) use time-sliced task labels, we use the sense-disambiguated corpus-frequency of a slang term across FineWeb dumps (Experiment 1). Slang is what makes this substitution possible: a fact flips from true to false, whereas a slang term traces a measurable lifecycle of rise and decline that allows us to evaluate slang usage's appropriateness (as a proxy for correctness) around a given time period.

2. **Facts that expire.** Zhang and Choi (2023) predict how long a fact stays true; our ephemerality criterion (Section 3) asks the

same question of a word, with the duration measured from the corpus. Unlike fact based evaluation, where an outdated fact is *wrong* and can be discarded, outdated slang remains grammatical and licit, so no training or evaluation objective penalizes it. Lexical misalignment is nuanced and missed by accuracy-based evaluations this literature relies on.

3. **Resetting the clock.** Zhao et al. (2024) show that a year cue in the prompt substantially realigns factual behavior. We run the direct lexical analogue (Experiment 3) and find that it does *not* recover recent vocabulary.Vocabulary appears to be less associated with specific time, as further demonstrated in Experiment 2.

4. **Surviving text per slice.** Cheng et al. (2024) treat corpus temporal leakage as a nuisance that corrupts the reported cutoff. Because a term's representation reflects total accumulated frequency rather than current relevance, a decade of accumulated 2010s text keeps decade-old slang dominant over terms that have only recently begun to rise.

In short, this body of work establishes that models are anchored in the past with respect to *what they know*. We show that they are anchored in the past with respect to *how they talk*, that this second anchoring survives the interventions that repair the first, and that it carries a socio-cultural cost that accuracy-oriented benchmarks do not register.

Related Work on Slang Processing and Understanding in LLMs Both Sun et al. (2024) and Wu and Sun (2025) are recent papers exploring LLM slang usage. Neither of these works examines the *temporal distribution* of slang usage, whether models favor older vs. newer slang, which is the central contribution of this paper. LLMs weaknesses in slang understanding and inability to generate slang in a way similar to human's, as demonstrated by the above papers, does provide additional support to our arguments that slang erasure is a real technical problem with models with real societal impact.

Model Collapse and LLM-Generated Data The risk of LLM-generated text contaminating future training corpora is increasingly well-documented Shumailov et al. (2023); Rahmani et al. (2025); Drayson et al. (2025); Feng et al. (2024). Shumailov et al. (2023) show that training on model-generated content causes irreversible defects where tails of the original distribution disappear.

This body of work motivates our central concern: if model outputs are temporally frozen and increasingly populate the web, future corpora will contain an ever-growing proportion of linguistically dated "synthetic" text. Even using detection to avert collapse (such as proposed by (Drayson et al., 2025)) the supply of fresh human-authored text is may shrink, leaving older data to dominate the high-quality fraction of training corpora.

3 Methodology

Dataset: FineWeb Our corpus baseline on slang trends is drawn from **FineWeb** (Penedo et al., 2024), a large, quality-filtered, deduplicated text dataset derived from Common Crawl and widely used to train open LLMs. We use the public 10BT sample on HuggingFace, whose documents retain a `dump` field identifying the originating CC-MAIN crawl (year-week). Dumps are available from 2013 to 2024, giving us a wide range to test for word ephemerality taking place on the scale of years or months.

Working from FineWeb rather than raw Common Crawl files has two advantages. First, it already applies the quality and filtering of modern training corpora, so our frequency estimates better reflect the text models actually train on than a raw Common Crawl would. Second, the pre-extracted sample is much cheaper to process than re-crawling and parsing the full web archives. Regarding the use of FineWeb as a proxy for internet usage trends overall, while FineWeb's own filtering and deduplication may distort raw frequencies, CommonCrawl's imperfect de-deduplication processes mean that it is not a perfect representation of real internet language usage either. We therefore treat the resulting rates as a representative proxy for slang prevalence, with high relevance to LLM training, in training data rather than an exact measurement of web usage.

Model Selection We run our query based vocabulary experiments on four models: `meta-llama/Llama-3.3-70B-Instruct-Turbo` and `deepseek-ai/DeepSeek-V4-Pro` provided via Together AI, and two closed models, `gpt-4o` (OpenAI) and `claude-sonnet-4-6` (Anthropic). Larger models show emergent capabilities that

smaller ones lack (Wei et al., 2022). This is why we experiment on recent, capable models rather than small ones alone: small-model results may not carry over, and whatever awareness of slang recency a model has is both more likely to exist and more likely to surface in its output at scale.

We test four models spanning three providers: two open-weight models served via Together AI, `meta-llama/Llama-3.3-70B-Instruct-Turbo` and `deepseek-ai/DeepSeek-V4-Pro`, and two closed models, `gpt-4o` (OpenAI) and `claude-sonnet-4-6` (Anthropic). Because the closed APIs do not expose token logprobs, responses are sampled (temperature=1.0) and summarized over samples by frequency.

Identifying Target Slang Words The main analysis is divided into two main methodologies, based on the source of the set of slang words in an experiment. The initial experiments involve identifying a reference set of slang words in the FineWeb corpus, and subsequently running analysis on their frequencies across FineWeb dumps to profile their temporal characteristics. We then examine how models use and view this set of corpus identified slang words. The second set of experiments starts by prompting a set of models with persona-style prompts intended to generate free-form slang usage. We use the responses to generate a list of slang commonly used by models, which we can then run analysis against in the FineWeb corpus. This gives us chracteristics of the slang words most commonly used by the set of models, such as peak usage age in the corpus for a given word. For the set of slang words used for initial analysis in FineWeb, we draw 125 candidates from Word-of-the-Year lists, Urban Dictionary, and Google Search Trends[2]. Each candidate is then **validated against the FineWeb corpus-frequency analysis of Section 4.1** using the quantitative *ephemerality* criterion below; for the experimental subset we additionally require that the word fill a recurring informal semantic slot (e.g. “impressive/cool”), keeping meaning roughly controlled across eras.

Quantitative ephemerality criterion. **TODO:** This should now be We definine "ephemerality" relatively loosely here and look at three types of words: 1) Words with rising popularity, peaking near the end of our dataset cutoff 2) Words with decreasing popularity, with a minimum near our dataset cutoff 3) Words with a peak somewhere within the bounds of the dataset. All three of these are filtered on words that have a >40% difference between the peak year usage in our dataset and the minimum year usage, and at least 80 instances across the dataset, to filter out extremely low-frequency noisy words.

Applying these criteria to the 125 candidates yields the "ephemeral" target set. The full evaluation across all 125 words can be found in B. The passing list of target words for subsequent experiments is the following:

1. **Pre-2018 (peaked early, now declining):** *lol, sick, troll, bro, swag, omg, meh, lmao.*

2. **Around 2020 (rising through the late 2010s):** *alpha, red pill.*

3. **2022–2024 (most recent, sharply rising):** *slay, gaslight, glow-up, aura, lowkey, situationship, vibe, vibes, legit.*

TODO: Drop alpha, vibe.... worst metrics by a good margin. Can bump up filter to 50% change **TODO:** Will now need to rerun all inference experiments with the expanded set (except the free generation ones) **TODO:** If we use second difference of raw data, we can check for "smootheness", in this case if we want to include all the above words and pick a cutoff then our cutoff value will be: <1.5 - drops red-pill. We also likely want to filter out some of the highly polysemous words. Potentially based on the roberta classifier metrics. Additional exclusions to consider: cancelled, slop, slaps, goon, fresh, yap, yapping, goals, ick, mid, karen, 6-7, sick, tight, twinning, u, pumped Additional words: pre-2018: finna, griefing, yap, goon, lmao, stan, hella, fresh, swag, yapping, rad, ick, karen, meh, 6-7, omg, troll, stfu, bro, sick, lol, tight, u, pumped, methinks, dude, lmfao, 2020: struggle bus, yas, unbox, cuck, cancelled, slop, shitposting, shitpost, cucked, goals, mid, twinning, bae, 2022-24: aura, hooman, slaps, catfished, skull-emoji, pog, vibing, glow-up, gaslight, lowkey, situationship, slay, yeet, legit, vibes, woke

Several of these words are polysemous (e.g. *sick*, *troll*, *aura*), with a slang sense distinct from a standard-English sense; Section 4.1 describes the sense-disambiguation step used to count only the slang usage.

[2]Words of the Year tend not to be slang per se, but instead often relate to global events. Using Urban Dictionary and cross referencing word popularity using Google Search Trends was the most reliable way to find good candidate words.

4 Analysis

4.1 Experiment 1: Corpus Frequency Over Time

4.1.1 Methodology

Goal. Establish a reference temporal frequency distribution of target slang words across time-sliced web data, serving as a ground truth against which LLM output distributions can be compared.

Data source. We use the **FineWeb** (Penedo et al., 2024) 10BT sample hosted on HuggingFace, which carries a `dump` field tagging each document with its originating CC-MAIN crawl (year-week), allowing us to slice the sample by crawl date[3]. FineWeb is itself a filtered, deduplicated derivative of Common Crawl, so it both reflects the kind of corpus used to train modern LLMs and removes much of the boilerplate and duplication that makes raw Common Crawl analysis difficult and unreliable. FineWeb processing provides data spanning 2013–2024, which matches with the published cutoff date for our open models used.

Sense disambiguation. A central difficulty is that many target words are polysemous: *sick* (ill vs. excellent), *troll* (creature vs. online provocateur), *aura* (energy field vs. Gen-Z coolness). To address disambiguating the slang and non-slang senses during counting, we use a methodology inspried by (Aloraini et al., 2026), we extract the textual context of every target-word occurrence and score it with a **RoBERTa classifier fine-tuned to distinguish the slang sense** from the standard sense. Only occurrences scoring above a threshold (≥ 0.99, selected to minimize false positives) are counted as slang usage. See Appendix D for full details of our evaluation of this schema.

Normalization. Sense-filtered counts are converted to a rate expressed as occurrences per million tokens of the corresponding dump:

$$r_{w,d} = \frac{\text{count}_{\text{slang}}(w,\, d)}{\text{tokens}(d)} \times 10^6$$

where w is a target word and d is a crawl dump. Per-dump token totals are taken from the FineWeb sample metadata. This normalization controls for the fact that successive dumps vary substantially in size.

[3]Crawl date's accuracy as a proxy for our true desired metric *published date* is reliant on the date sources effective deduplication, i.e. how well any given dump does excluding old material not introduced since the last checkpoint.

Uncertainty. The 10BT sample is a random draw from FineWeb, so a per-dump occurrence count k is modeled as $k \sim \text{Poisson}(r_{w,d} \cdot \text{tokens}(d))$. We report the exact Poisson interval (95% by default) as a shaded band around each word's rate curve; bands are wide for rare words and tight for common ones.

Output. For each target word we plot $r_{w,d}$ against dump date, producing the temporal frequency curves in Section 4.1. These curves are the corpus ground truth against which the LLM generated word distributions are compared.

4.1.2 Findings

Slicing the FineWeb 10BT sample by crawl dump and applying the RoBERTa sense filter yields, for each target word, a curve of slang-sense occurrences per million tokens against dump date (2013–2024). The band figures below plot each curve as a percentage of that word's own peak dump, so words spanning two orders of magnitude in absolute rate can be compared by shape on a shared axis; the absolute peak rates are reported in Table 1. These curves confirm that the target words exhibit the ephemeral lifecycles the querying experiments rely on, and they separate cleanly into the three temporal bands of Section 4.1.1, which we present in Figure 1.

Two properties matter for the following response vocabulary analysis. First, the words we treat as "older" (*lol*, *swag*) are demonstrably in decline by the end of the window, while the "recent" words (*aura*, *slay*, *glow-up*) are still rising. A true-to-life model of slang usage would emphasize using more recent and relevant slang words, while deemphasizing older slang that has become dated and much less popular. Second, the sense filter is what makes these counts meaningful: without it, *sick* and *aura* would be dominated by their standard-English senses and show no slang lifecycle at all. Additional details regarding the sense filtering process, such as model recall and precision, are included in the appendix.

4.2 Experiment 2: Direct Year Association

4.2.1 Methodology

Goal: Experiment 3 measures *usage*, essentially what slang the model uses when it is not told to think about time. Experiment 2 instead measures *meta-knowledge*: when asked directly, does the model know when a slang word was popular? Contrasting the two isolates whether failures are a

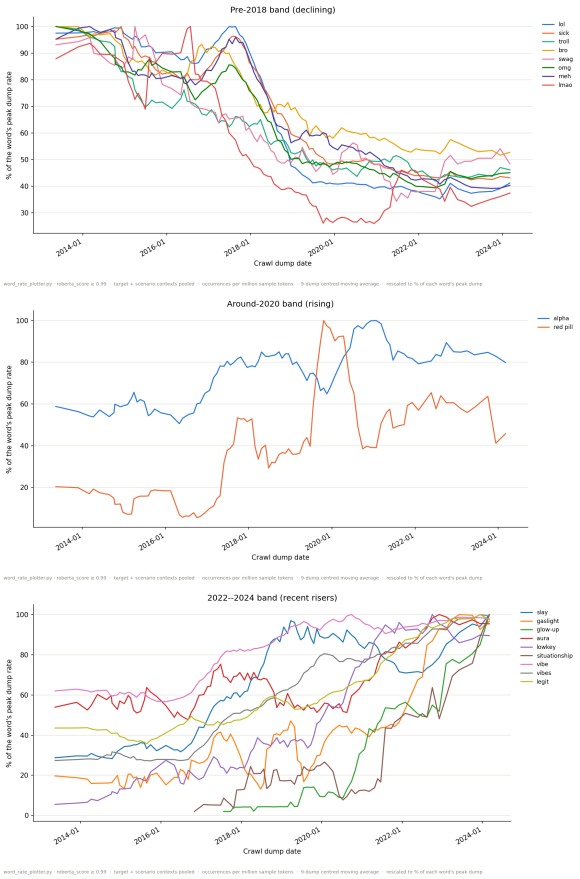


Figure 1: Slang-sense frequency by FineWeb crawl dump with each word plotted as a percentage of its own peak dump. Rescaling lets terms two orders of magnitude apart in absolute rate (*lol* peaks near 16 occurrences per million, *swag* below 0.5) share one axis. Generated with `word_rate_plotter.py -threshold 0.99 -highlight-bands -percent-of-peak -no-ci -clean -settings-note -smooth 9`: rows scoring $\geq$ 0.99 from the pooled target and scenario scored contexts, normalised to occurrences per million tokens of the corresponding FineWeb 10BT sample dump, smoothed with a 9-dump centred moving average; dumps with no occurrences are dropped rather than plotted at zero. Word selection: every word plotted passes the ephemerality criterion of Section 3 — at least 100 slang-sense hits (or 50 with an unmistakable trend, $R^2 \geq 0.8$ and $|\beta| \geq 0.3$ bits/yr) together with a non-flat trajectory, and is assigned to the band matching its corpus peak year. Absolute peak rates and per-word peak years are given in Table 1.

knowledge gap or a usage bias. This is the inverse of the Experiment 3 design constraint: here we deliberately provide a temporal cue via the prompt.

We use three prompt forms, each rendered once per target word and sampled across all four models:

1. **GiveYear:** *"The slang word '{word}' is most associated with what year? Respond using only the year."*: directly requesting the word's peak year.

2. **DatePhrase:** *"You hear someone saying '{word}' in conversation, which year are they most likely saying this in?"*: a softer, scenario-framed variant of the same question.

3. **EraAppropriate:** *"It's {year}, is it appropriate to use the term '{word}'?"*: a year-conditioned appropriateness judgement swept over 2013–2025. Rather than asking the model to *name* a year, this form fixes the year and asks the model to *judge* whether the term fits it, testing whether meta-knowledge of slang timing surfaces as a usage judgement.

GiveYear and DatePhrase are sampled five times per word per model, and we parse a four-digit year from each free-text response; comparing the model's stated peak year against the corpus peak from Experiment 1 tells us whether a model's *declarative* knowledge of slang timing is accurate. EraAppropriate is sampled five times per (word, year) pair and each answer is classified yes/no — if a model tracked usage, appropriateness should fall off after a word's corpus peak. The full prompt set (including supplementary logprob and inverse-mapping prompts) and per-prompt counts are given in Appendix E.

4.2.2 Findings

Experiment 2 asks what the models *know* about slang timing, by posing the temporal question directly. We see that while models carry *directionally* correct declarative knowledge of when slang was popular, that knowledge is both noticeably shifted toward the past and not reflected in their spontaneous usage (Experiment 3).

The figure overlays model belief against corpus ground truth. Figure 2 places the target words on the y-axis *ordered by their true corpus peak year* (from Experiment 1), with calendar year on the x-axis. Because the words are sorted by true peak, the stars form a rising staircase, so the horizontal

offset of each diamond from its star reads directly as how far the model is off.

Models order slang roughly by era but compress it toward the past. The diamonds broadly follow the staircase — models do place *lol*, *troll*, and *bro* early and *aura* and *slay* late — but they systematically date the newer terms too early. For `claude-sonnet-4-6`, *alpha* (corpus peak 2020) is placed near 2010, and *vibe*/*vibes* (still rising through 2024) near 2018/2020. The mean absolute error against the corpus peak is sizeable for every model: 6.5 years for Claude, 7.1 for Llama, and 10.3 for DeepSeek, consistent with a sense of slang anchored to an earlier window than current usage.

Caveat on the comparison. The corpus "peak" is the year of maximum frequency, so for the words still rising at the end of the data window (2023–2024 peaks) it is closer to a lower bound on recency; a model dating *vibe* to 2018 is arguably reporting the *onset* of popular use rather than the corpus peak. The comparison is cleanest for words that have visibly crested within the window (*lol*, *swag*, *alpha*), where the models still skew early. `gpt-4o` is a special case: its lower 5.5-year error is an artifact rather than better calibration, since its *DatePhrase* responses collapse to "2023" (close to its cutoff date) for almost every word, which happens to sit near the many recent peaks.

Asked to judge rather than name a year, models never call slang dated. A complementary query set (*EraAppropriate*, Appendix E.1) fixes the year in the prompt and asks whether a term is appropriate to use then. If the directional timing knowledge in Figure 2 were carried through to a usage judgement, appropriateness would fall off after a word's corpus peak. It does not: across 2013–2025 the models answer "appropriate" for almost every (word, year) pair (yes-rate 100% for `gpt-4o`, 96.7% for Claude, 88.4% for DeepSeek, 78.6% for Llama), and for no word does the yes-rate decline after the peak except under `Llama-3.3-70B`, the only model showing even a partial drop-off ($84\% \rightarrow 72\%$). Even the era knowledge the models demonstrably have thus fails to surface when the same fact is queried as an appropriateness judgement.

Interpretation. The models possess *directional* temporal knowledge but that knowledge is both compressed toward the past and, critically, not carried through in their actual usage (Experiment 3). The failure to produce current slang is therefore

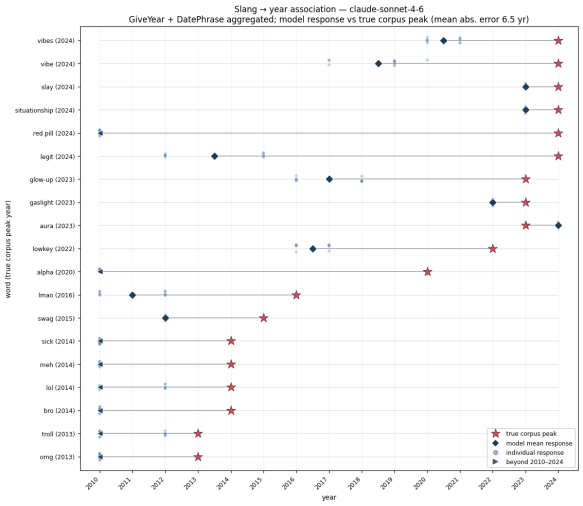


Figure 2: Direct year association for `claude-sonnet-4-6`, pooling the *GiveYear* and *DatePhrase* prompts. Words (y-axis) are ordered by their true corpus peak year (Experiment 1, shown in parentheses); the x-axis is calendar year. For each word the red star is the corpus peak, the dark diamond is the model's mean responded year, the connecting line is the error, and faint dots are individual responses; markers falling outside 2010–2024 are drawn as arrows at the axis edge. The diamonds track the rising staircase of true peaks only loosely and skew early for the newer terms (mean absolute error 6.5 years). Per-model figures for `gpt-4o` (5.5 yr, inflated by its collapse to "2023"), `DeepSeek-V4-Pro` (10.3 yr), and `Llama-3.3-70B` (7.1 yr) are included as supplementary figures (`give_year_date_phrase_*.png`).

not a total absence of temporal knowledge, but shows the divergence between model's declarative knowledge and behavior, a particularly important discrepancy for evaluting how models may interact with language trends.

4.3 Experiment 3: Semantic Slot Querying

4.3.1 Methodology

Goal: Measure what slang a model actually *produces* when asked for an informal word, with no temporal cue in the prompt, so we observe natural usage rather than meta-knowledge about eras. A model whose vocabulary tracked current usage should readily produce recent slang; a temporally frozen model should fall back on older, well-attested terms. A central design constraint is that we avoid naming years or providing informal language queues in the prompt, which would test meta-knowledge or register-matching instead of general usage.

We give the model a temporally neutral semantic description of a meaning and ask for a single informal word. The test runs in two modes:

Concept-slot mode. The description names a general meaning with several era-spanning valid answers ("something impressive/cool", "mediocre", "charisma", "suspicious", . . .) and we record which word the model reaches for, surfacing its default vocabulary per slot without naming any target word.

Target-recovery mode. The description is written to point at one specific target word from Section 4.1.1 — e.g. a definition that uniquely evokes *gaslight* or *aura* — and we measure how often the model returns that exact word, counting stem and inflection matches (so *glow-up*/*glowup* and *gaslight*/*gaslighting* all count). Recovery rate is a direct measure of how reluctant the model is to produce the intended term, and of whether the term is effectively out of vocabulary. Each target word is queried for with three independently worded phrasings to control for wording, and each phrasing is sampled ten times per model.

Year-conditioned variant. To test whether a model conditions its vocabulary on a stated time, we prepend "The year is {year}." to each target-recovery prompt and sweep {year} over 2013–2025. If a model were aware of its own data distribution, we would expect recovery accuracy for a word should be highest at, or near, the year of that word's true corpus peak (Experiment 1). Flat accuracy across years would indicate the model ignores the temporal cue; a peak at the wrong year would indicate a miscalibrated sense of timing, or one based on something other than dataset distribution.

4.3.2 Findings

This experiment is our first framework for querying models to explore their awareness of slang's temporal characteristics. We generate a selection of prompts that are intended to solicit one of a set of target words from the models, and test their ability to predict the intended word. The list of target words is the set of slang words from section one, with known temporal characteristics. We run two variations of this experiment: One with a free response from the model from which target words are identified, and a second where the model chooses from a multiple choice set with the target slang term as one of the options.

Target-word recovery **TODO:** Fill in and rewrite this whole section It seems like for words where we had close synonyms as options in the multiple choice, we actually did see slight spikes in the correct choice based on year (omg, sick, aura). Others tended to max stay at around 100 percent, likely due to no other good options being provided in the multiple choice.

Conditioning on a stated year The year-conditioned variant prepends "The year is {year}." to each recovery prompt and sweeps the year over 2013–2025. If a model tracked its own data distribution, recovery for a word should be highest at that word's true corpus peak year. Figure 4 shows the result as a word × prompt-year accuracy heatmap for `claude-sonnet-4-6`, with rows ordered by true corpus peak and each word's true peak year outlined.

The stated year has little effect. For the majority of words accuracy is essentially flat (near 100%) across all thirteen prompt years (See Experiment 3) **TODO:** Add subexperiment indices? like i, ii, etc.. Changing "2013" to "2024" leaves the output unchanged. The model is not conditioning its vocabulary on the stated time, even when the stated year is one that has fallen out of common use.

Where accuracy does vary, it does not track the true peak. Among the words with year-to-year variation, the high-accuracy cells do not line up with the outlined true-peak cell. The starkest case is `gpt-4o`'s *red pill* (true corpus peak 2024): recov-

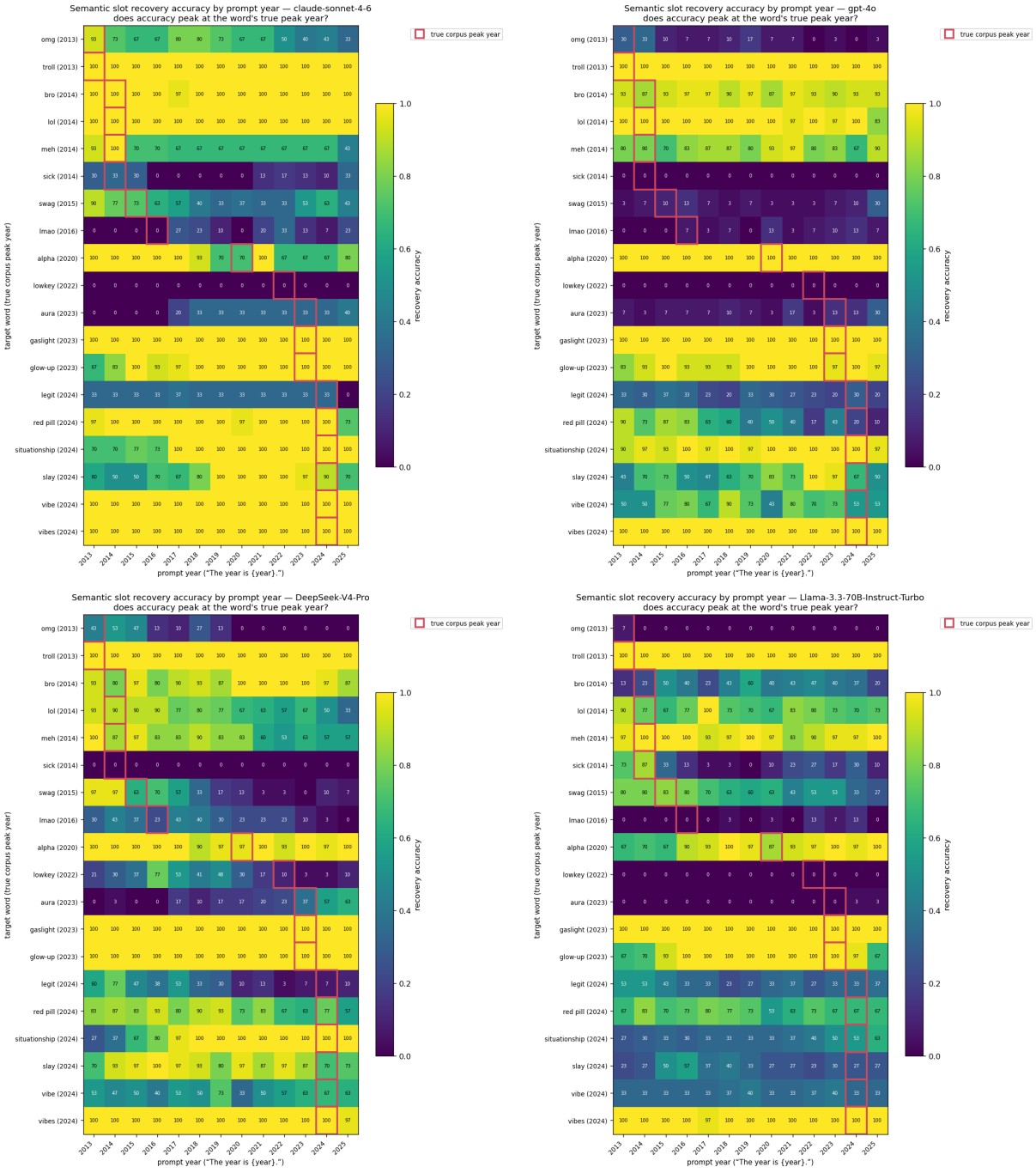


Figure 3: Semantic-slot target-word recovery accuracy for all four models: `claude-sonnet-4-6` and `gpt-4o` (top row), `DeepSeek-V4-Pro` and `Llama-3.3-70B` (bottom row). Within each panel, a cell is the recovery accuracy for a target word (y-axis, ordered by true corpus peak year, shown in parentheses) when the prompt is prefixed "The year is {year}." for the year on the x-axis (30 samples per cell; stem and inflection matches counted), and each word's true corpus peak year is outlined. Reading down a column shows which words a model can produce; reading across a row shows the (near-absent) effect of the stated year. Older, unambiguous terms are recovered reliably across all years, while polysemous (*sick*, *swag*), subtle (*lowkey*), and very recent (*aura*) terms are recovered poorly — a pattern that holds across all four models.

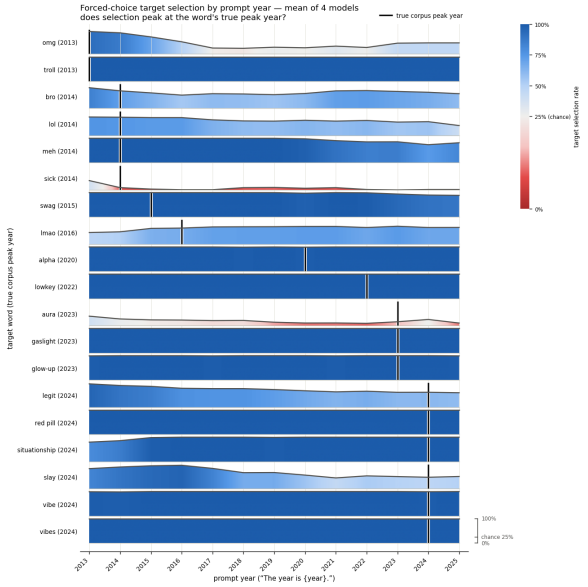


Figure 4: Multiple choice slang test results: Aggregate results for asking all four models to select the most appropriate word/phrase from 4 multiple choice options. The models remain highly confident of their choice in most cases even if the year in the prompt shows low usage of the word in the actual corpus. This indicates that the models do not reason about the temporal relevance of a slang word of phrase on generation.

ery is *highest* in the earliest prompt years (∼90% at 2013) and *lowest* at the true peak (20% at 2024). Claude's *alpha* (peak 2020) is recovered at 100% through 2013–2017 but dips to ∼67% around its actual peak. If anything, models recover a word slightly better when told an *earlier* year, which matches the backward-shifted sense of timing quantified in Experiment 2.

Taken together, the two views give a consistent reading: models reliably produce older, well-attested slang but struggle with the most recent terms, and an explicit year cue does not shift them toward era-appropriate vocabulary.

4.4 Experiment 4: Persona Prompting

4.4.1 Methodology

Goal: The other queries all pin the model down to a single-word answer. Experiment 4 attempts to invert the goal of querying information about known slang words from the models, instead seeking to find what slang is commonly generated by models and then analyzing the characteristics of that set of words.

Prompts. We give the model a persona instead of a question. The system prompt tells it to act like "a young person using informal language like you would among peers," and two user prompts ask it to simulate a short text-message or group-chat exchange between friends. The prompt never names a target word, a year, or even the word "slang," so whatever informal vocabulary shows up is volunteered rather than elicited. We collect 100 free-form responses per model (400 in total) at temperature 1.0.

Scoring via effective vintage. To turn free text into a temporal measurement, we scan each response for words from a 150-item slang lexicon (generated by combining the words from section 1 with manually extracted phrases/variants from this experiment's responeses) and tag every hit with that word's corpus peak year [4]. We then summarise a model's output by its *effective vintage*: the occurrence-weighted mean peak year of the slang it produced. A model's datedness can be measured by how far behind present day it's *effective vintage* lies. As a baseline we compute the same weighted mean over the lexicon's corpus frequencies (its *corpus vintage*), so the reference point is the corpus's own centre of mass rather than the calendar, since both are expected to show some offset from present day. This frame of reference serves to confirm that the model does not exhibit emergent behavior that might bring it's slang usage closer to current trends. Appendix H gives the estimator, its bootstrap confidence intervals, and a distributional cross-check.

4.4.2 Findings

Experiment 4 inverts the direction of investigation, starting with relatively free form "persona" generations by the models, and analysing the generated vocabularies temporal properties. Figure 5 counts how often each lexicon word appears across the 400 persona responses, ordered by corpus peak year (top panel) and by corpus frequency (bottom). A few long-lived, high-frequency terms dominate: *u*, *lol*, and *omg* together make up roughly 79% of all in-lexicon slang, and adding *bro* brings the four terms that peaked in 2013–2014 to about 84% of the total. Recent slang, such as *vibe*, *slay*, *legit*, *aura*, turn up in the low single digits or not at all, echoing the recovery gap from the semantic-slot queries.

The output carries an early-2010s vintage. Putting a number on the skew, the slang these mod-

[4] Peak year is calculated using the word counts generated in section 1. For words added to the set from the experiment 4 responses we re-ran an identical set of steps to initial slang collection from section 1 to find words counts per dump.

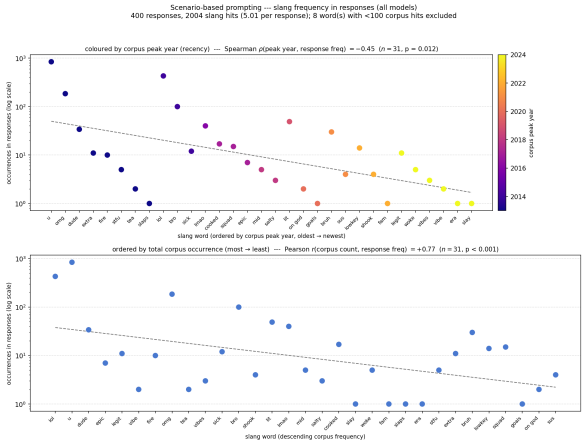


Figure 5: Slang word frequency in naturalistic persona-simulation responses (log scale), aggregated across four models (400 responses, 100 per model). Top panel: words ordered by corpus peak year (oldest → newest), coloured by peak year (plasma scale). Bottom panel: same words ordered by descending corpus frequency. Dashed lines are log-linear trend fits. High-frequency, older terms (*u*, *lol*, *omg*) dominate both panels; recent slang appears at the far right of the top panel with counts near 1, reproducing in free generation the recency gap observed in forced-choice recovery.

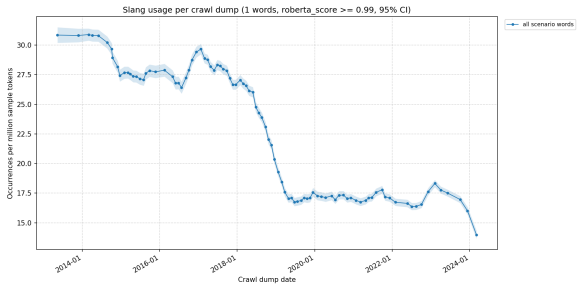


Figure 6: Corpus baseline for the persona lexicon: aggregate slang-sense frequency across the full scenario vocabulary in FineWeb 10BT, normalized per million tokens per crawl dump and summed across all words. The aggregate curve rises through the mid-2010s and plateaus, reflecting the combined lifecycle of the informal lexicon the models draw on rather than any single word's arc. This is the corpus-side reference whose centre of mass (a *corpus vintage* of 2015.8) the models' 2014.0 output vintage is measured against.

els emit has an occurrence-weighted effective vintage of **2014.0** (95% CI [2013.9, 2014.1]; Poisson bootstrap), roughly **1.7 years older** than the **2015.8** vintage of the same lexicon in the corpus (Figure 6). Every model skews old, so this is not a single outlier dragging the average down. We do not observe any emergent bias in the models to generate more recent slang words to better fit with trends in language. Instead we unsurprisingly see a relatively close match to the distribution in the training data, with a tendency to overuse phrases that show recent decreases in popularity (lol, omg).

Reluctance versus recency. The models differ mainly in how much slang they volunteer at all: `gpt-4o` produces the fewest in-lexicon hits of the four (205, against 881 for `claude-sonnet-4-6`), in keeping with its general reluctance to slip into a casual register unprompted (). What none of the four do is reach for current slang; the early-2010s skew holds across all of them.

5 Results

Findings from the corpus analysis and the query experiments converge on a consistent pattern:

1. **The target words have genuine, separable lifecycles.** The corpus analysis (Experiment 1, Figure 1) confirm that the older terms (*lol*, *swag*) are in decline by the end of the window while the newest terms (*aura*, *slay*, *glow-up*) are still rising. This is what makes the words usable as a temporal probe and informs the curated target set selection.

2. **Models are least fluent in the most current slang.** In semantic-slot recovery (Experiment 3, Figure 3), every model recovers older, unambiguous terms (*lol*, *troll*, *gaslight*, *glow-up*) near-perfectly, but recovery collapses for the newest term *aura* and for polysemous slots (*sick*, *swag*) where the model prefers a competing or literal word. The slang that is most current in human usage is the slang the models are least able to produce on demand.

3. **An explicit year cue does not fix it.** Prefixing the recovery prompt with "The year is {year}." (Experiment 3, Figure 4) barely changes which slang the model produces: most words are recovered at a flat rate across 2013–2025, and the words that do vary do not peak at their true corpus-peak year — for the most recent terms (*red pill*, *vibe*) accuracy is, if anything, highest when an *earlier* year is named. The stated date does not seem to influence the models' confidence in choosing out-of-date slang terms.

4. **Parametric knowledge vs usage are unaligned.** The year-association queries (Experiment 2, Figure 2) show that models order slang by era directionally correctly, but

place the newer terms years too early relative to the corpus peak (mean absolute error 5.5–10.3 years per model), and their spontaneous usage does not track recency at all. The temporal failures contain both a usage bias and a backward-shifted sense of slang timing.

5. **Freeform usage patterns reflect bias against modern slang.** When the models are prompted to write casually in a persona (Experiment 4, Figure 5), the slang they volunteer still skews old. Occurrence-weighted effective vintage (as outlined in Experiment 4's methodology section) is 2014.0, about 1.7 years behind the 2015.8 vintage of the same lexicon in the corpus, and a handful of long-lived terms (*u*, *lol*, *omg*, *bro*) account for the bulk of usage.

6 Discussion

6.1 Interpretation

Results suggest the models are not sensitive to temporal trends in their spontaneous slang usage. When asked to produce current slang on demand they reliably return older, well-attested terms and struggle most with the newest ones (Experiment 3). An explicit year cue does not move them toward era-appropriate vocabulary, even while they retain *directionally* correct declarative knowledge of when those terms were popular (Experiment 2). This is consistent with the hypothesis that model slang *usage* reflects total frequency in the training corpus rather than a temporally calibrated view of what is current. While this may be an intuitive result, we argue that the implications of this quality of LLMs has severe downstream implications for the model ecosystem and its societal impacts. **TODO:** Strengthen the framing with linguistics citations on the social and identity-signaling function of slang, motivating why temporally miscalibrated informal vocabulary matters independently of the downstream training-data argument.

The difficulty models have producing the most recent term *aura* (which is still sharply rising in the corpus at the end of the window in mid 2024) is striking precisely because the same models date it to roughly the right era (2023–2024) when asked directly (Figure 2). *aura* is the cleanest illustration of usage-bias-without-a-knowledge-gap: the model knows when the word belongs yet underrepesents it in generation. For most other recent terms the model additionally shifts its stated date years too early, so there is unclear declarative knowledge in addition to miscalibrated usage in generation. The reliability gap between recovering older versus newer slang may reflect that the older terms have saturated the training data with strong, redundant signal, whereas the newest terms appear in quantities too small (and too close to the data-collection horizon) to anchor confident generation, despite being the dominant form in current human usage.

6.2 Implications for Training Data

If models cannot keep pace with linguistic change, the effect compounds as web text becomes increasingly model-generated (Shumailov et al., 2023): output reflecting stagnant, older vocabulary reenters future training corpora, further entrenching dated language and eroding models' capacity to represent current human usage. This connects to the "stochastic parrots" critique (Bender et al., 2021)—models capture the statistical structure of language but need not track its dynamics, leading to a fundamentally inaccurate view of human language.

The practical effect today is modest, and is partly masked by instruction-tuned models' general reluctance to produce slang at all (Section 4.3) (which we would argue has further social implications as young people get a higher and higher percentage of their linguistic contact from models). The longer-term concern is both larger and more general than slang: that models lagging ability to reflect current usage trends makes them a perpetually outdated interface, and inaccurate representations of language (or by extension personas). As models are retrained on the accumulated text of past decades—increasingly including their own outputs—the language they produce may drift steadily further from contemporary usage. Mitigating this will require explicit training interventions, potentially things like recency-weighted sampling, together with a sustained supply of fresh, verifiably human-authored text. Our results add weight to the case for treating human-generated text as a premium training resource critical to maintaining models that use language accurately.

7 Limitations

1. Cross-model comparisons confound knowledge cutoff with architecture, training data, and alignment procedure (e.g. RLHF); we therefore emphasize within-model temporal patterns over cross-model magnitudes.

2. All four models experimented on are instruction-tuned. RLHF and instruction tuning are known to suppress non-standard English (Barnhart et al., 2025) and may amplify the apparent temporal freeze. Replicating on base models would isolate this effect, but would limit us to experimenting on open models instead of more popular closed ones.

3. Quality filtering and deduplication in Common Crawl derivatives can distort raw term frequencies; our FineWeb-based rates inherit any such distortion, and the de-facto upsampling of older dumps (Cheng et al., 2024) biases the corpus baseline toward earlier usage. Mitigation of this is sought by the FineWeb filtering methodologies. Additionally, the conclusions here are concerned with the pragmatic effects of the training data on model language, so distortions in training data do not detract from the conclusions.

8 Future Work

Mitigation Suggested by Related Work From related work around staleness of models with regard to factual information there are several suggested mitigations that could apply here. Zhang and Choi (2023) model how long a fact remains true and discard facts whose predicted duration has expired, trading recall for reduced staleness in open-domain QA. Zhao et al. (2024) show that a pretrained model's internal sense of "now" lags its actual data, and that this internal clock can be *reset* by conditioning on the current year in the prompt (which we demonstrate does not hold for the case of slang vocabulary correction), or by lightweight finetuning — recovering a substantial share of the lost performance. Exploring how finetuning might improve vocabulary used by a model is not explored in this paper and would be an interesting future set of experiments. Additionally, we suggest that current RLHF also has a strong effect on language style used by models **TODO:** citations. Correctly using RLHF from annotators that are familiar with the correct use of modern langauge could also improve the ability of models to correctly prefer up-to-date language.

Further directions.

1. The effects of RLHF are suspected to be a strong factor is current model vocabularies. Exploring these effects could involve replicating the queries on base (non-instruction-tuned) checkpoints to isolate the contribution of RLHF to slang suppression.

2. In coming years there will be a longer and longer history of models to study their evolution over time. To emulate a version of this we could run a longitudinal study tracking vocabulary drift as models are trained on successive crawl snapshots, which is unfortunately outside of the resource scope of this project (as it requires repeated, expensive pretraining).

3. Multilingual analysis: low-resource languages may be affected more strongly (model output can quickly dominate scarce human data) or less (smaller corpora may be more easily refreshed by recent text).

Acknowledgments

TODO: Fill in acknowledgments before camera-ready submission.

References

Oshin Agarwal and Ani Nenkova. 2022. Temporal effects on pre-trained models for language processing tasks. *Transactions of the Association for Computational Linguistics*, 10:904–921.

Afnan Mohammed Aloraini, Riza Batista-Navarro, Goran Nenadic, and Viktor Schlegel. 2026. The SlangTrack dataset: Supporting the detection of words used in slang senses. In *Proceedings of the 6th International Workshop on Computational Approaches to Language Change (LChange'26)*, Manchester, UK. Association for Computational Linguistics.

Stefan Baack. 2024. A critical analysis of the largest source for generative AI training data: Common Crawl. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*.

Logan Barnhart, Reza Akbarian Bafghi, Stephen Becker, and Maziar Raissi. 2025. Aligning to what? Limits to RLHF based alignment. *arXiv preprint arXiv:2503.09025*.

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623.

Jeffrey Cheng, Marc Marone, Orion Weller, Dawn Lawrie, Daniel Khashabi, and Benjamin Van Durme.

2024. Dated data: Tracing knowledge cutoffs in large language models.

Common Crawl Foundation. 2007. Common crawl — open repository of web crawl data. https://data.commoncrawl.org/crawl-data/index.html. Accessed: 2025.

George Drayson, Emine Yilmaz, and Vasileios Lampos. 2025. Machine-generated text detection prevents language model collapse.

Bethany K. Dumas and Jonathan Lighter. 1978. Is slang a word for linguists? *American Speech 53, no. 1.* 5–17. https://doi.org/10.2307/455336.

Kim Earl. 1972. Semantic influence and concept attainment of slang and its effects on parents' and teenagers' linguistic interaction. *All Graduate Theses and Dissertations, Spring 1920 to Summer 2023. 2637.* Https://digitalcommons.usu.edu/etd/2637.

Connie Eble. 1989. *Slang and Sociability: In-Group Language among College Students*. University of North Carolina Press. TODO: Verify edition and page numbers.

Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. 2024. Beyond model collapse: Scaling up with synthesized data requires verification.

H. W. Fowler. 1926. *A Dictionary of Modern English Usage*. Oxford University Press, Oxford.

William L. Hamilton, Jure Leskovec, and Dan Jurafsky. 2016. Diachronic word embeddings reveal statistical laws of semantic change. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1489–1501, Berlin, Germany. Association for Computational Linguistics.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, and 1 others. 2022. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*.

Barbara Kipfer and Robert Chapman. 2010. *Dictionary of American Slang*. Harper Collins. TODO: Verify edition and page numbers.

Alexandra Luccioni and Joseph Viviano. 2021. What's in the box? a preliminary analysis of undesirable content in the common crawl corpus. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pages 182–189.

Kelvin Luu, Daniel Khashabi, Suchin Gururangan, Karishma Mandyam, and Noah A. Smith. 2022. Time waits for no one! analysis and challenges of temporal misalignment. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5944–5958, Seattle, United States. Association for Computational Linguistics.

Allan A. Metcalf. 2002. *Predicting New Words: The Secrets of Their Success*. Houghton Mifflin, Boston.

Guilherme Penedo, Hynek Kydlíček, Loubna Ben allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. 2024. The fineweb datasets: Decanting the web for the finest text data at scale. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Hossein A. Rahmani, Varsha Ramineni, Emine Yilmaz, Nick Craswell, and Bhaskar Mitra. 2025. Towards understanding bias in synthetic data for evaluation.

David Reinsel, John Gantz, and John Rydning. 2018. The digitization of the world from edge to core. Technical report, International Data Corporation.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2023. The curse of recursion: Training on generated data makes models forget.

Zhewei Sun, Qian Hu, Rahul Gupta, Richard Zemel, and Yang Xu. 2024. Toward informal language processing: Knowledge of slang in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.

Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. 2024. Will we run out of data? limits of llm scaling based on human-generated data. *In Forty-first International Conference on Machine Learning*.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, and Donald et al. Metzler. 2022. Emergent abilities of large language models. *Transactions on Machine Learning Research*.

Siyang Wu and Zhewei Sun. 2025. How do language models generate slang: A systematic comparison between human and machine-generated slang usages.

Michael J.Q. Zhang and Eunsol Choi. 2023. Mitigating temporal misalignment by discarding outdated facts. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14213–14226, Singapore. Association for Computational Linguistics.

Bowen Zhao, Zander Brumbaugh, Yizhong Wang, Hannaneh Hajishirzi, and Noah A. Smith. 2024. Set the clock: Temporal alignment of pretrained language models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, and 3 others. 2023. A survey of large language models. *arXiv preprint arXiv:2303.18223*.

A Target Words and Corpus Peaks

Table 1 lists the nineteen tested slang terms, the temporal band they belong to, and their true corpus peak year — the calendar year of maximum slang-sense frequency in the FineWeb 10BT sample (Experiment 1, RoBERTa sense score ≥ 0.99). The band names the era in which a term rose to prominence; the peak year is the corpus argmax. The 10BT sample ends at dump `2024-10`, so for terms still rising at that horizon (*vibe*, *vibes*, *red pill*, *slay*) the listed peak is the final available year and should be read as a lower bound on recency.

Word	Band	Peak	Peak rate	Hits
troll	pre-2018	2013	1.008	6,077
bro	pre-2018	2014	1.043	7,914
lol	pre-2018	2014	16.162	107,305
sick	pre-2018	2014	2.017	14,015
swag	pre-2018	2015	0.408	2,624
omg	pre-2018	2013	4.125	24,593
meh	pre-2018	2014	0.844	5,775
lmao	pre-2018	2016	0.716	3,816
alpha	around 2020	2020	0.702	5,764
red pill	around 2020	2024	0.135	546
gaslight	2022–2024	2023	0.072	316
glow-up	2022–2024	2023	0.041	98
aura	2022–2024	2023	0.178	1,231
slay	2022–2024	2024	0.308	1,795
vibe	2022–2024	2024	3.878	30,862
vibes	2022–2024	2024	4.003	20,311
lowkey	2022–2024	2022	0.114	567
situationship	2022–2024	2024	0.077	121
legit	2022–2024	2024	5.350	32,897

Table 1: Tested target words, temporal band, corpus peak year, peak-year slang-sense rate (occurrences per million tokens), and total slang-sense hits across the FineWeb 10BT sample (RoBERTa sense score ≥ 0.99). Peak year is the argmax, over calendar years, of the token-normalised yearly rate; "Hits" is the raw count of above-threshold occurrences pooled over all 95 dumps. Rates span nearly three orders of magnitude, from the saturated *lol* (16.2 per M) to the emergent *glow-up* (0.04 per M).

TODO: Replace this whole section with a percentage based criterion I hope. Maybe keep a min threshold as well.

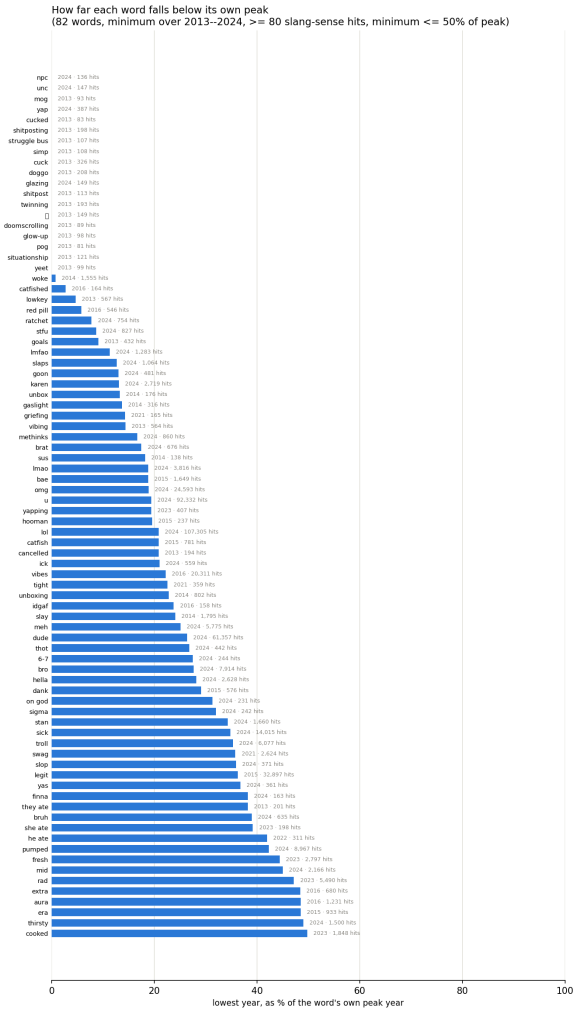


Figure 7: Chart of the subset of the 125 initially collected candidate words that pass the following filtering criteria: Change of >50% between maximum frequency in the dataset and the minimum year. Total count of >80 across the dataset, to exclude highly infrequent, noisy words. 82 total words. Generated using `python3 plot_min_pct.py -min-hits 80 -max-min-pct 50`.

B Word Selection Criteria and Statistics

B.1 Trajectory smoothness: the normalised second difference

The percent-of-peak screen above scores each word on a single pair of years—its peak and its trough—and is therefore indifferent to the path between them. A word that rises once and falls once and a word that oscillates across the whole decade can report the same drop. To separate the two we score the *shape* of each trajectory with a discrete roughness statistic: the root-mean-square second difference of the series, normalised by the series' own standard deviation.

Write x_t for a word's slang-sense rate in calendar

year t expressed as a percentage of that word's own peak year, so x is a series of $n = 12$ values (2013–2024) whose maximum is exactly 100. The roughness is

$$\rho = \frac{\mathrm{RMS}(\Delta^2 x)}{\mathrm{SD}(x)} = \frac{\sqrt{\dfrac{1}{n-2}\displaystyle\sum_{t=2}^{n-1}(x_{t+1} - 2x_t + x_{t-1})^2}}{\sqrt{\dfrac{1}{n-1}\displaystyle\sum_{t=1}^{n}(x_t - \bar{x})^2}}. \quad (1)$$

Explicitly, for each word:

1. **Build the series.** Divide each year's token-normalised rate by the word's peak-year rate and multiply by 100, giving twelve values with the peak pinned at 100.
2. **First difference.** $\Delta x_t = x_t - x_{t-1}$, eleven values: the year-on-year change, i.e. the local slope.
3. **Second difference.** $\Delta^2 x_t = x_{t+1} - 2x_t + x_{t-1}$, ten values: the change *in* the slope, i.e. how sharply the line bends at each interior year. This is the discrete analogue of f'', and it is why ρ is trend-immune: a straight line of any steepness has constant slope, so every second difference is exactly zero and only departures from linearity are counted.
4. **Root-mean-square.** Square the ten second differences, average, take the square root. Squaring prevents an upward bend and a downward bend from cancelling; the result is the typical bend size in percentage points.
5. **Normalise.** Divide by the sample standard deviation of x ($n - 1$ denominator). Without this step a word ranging over eighty points would outscore one ranging over ten purely because its units are larger; the ratio instead asks how large a typical bend is relative to how much the series moves overall.

Worked example. Table 2 carries out the calculation for *aura*. Summing the squared second differences gives $\sum(\Delta^2 x)^2 = 5280.9$; dividing by ten and taking the root gives $\mathrm{RMS} = 22.98$, and the series' standard deviation is 17.10, so $\rho = 22.98/17.10 = 1.34$. The sign of the second difference alternates almost every year, which is what a large ρ records. For contrast, *vibes* travels further overall ($\mathrm{SD} = 25.72$) but climbs steadily, so its bends are small ($\mathrm{RMS} = 7.57$) and $\rho = 0.29$—nine times smoother on identical arithmetic.

Year	x_t	Δx_t	$\Delta^2 x_t$
2013	60.7		
2014	58.3	−2.4	
2015	53.0	−5.3	−2.9
2016	48.5	−4.5	+0.8
2017	69.2	+20.7	+25.2
2018	66.9	−2.3	−23.0
2019	51.8	−15.1	−12.8
2020	51.4	−0.4	+14.7
2021	78.8	+27.4	+27.8
2022	93.7	+14.9	−12.5
2023	100.0	+6.3	−8.6
2024	54.1	−45.9	−52.2

Table 2: The roughness calculation for *aura*: percent of peak, first difference, and second difference. The ten second differences yield $\mathrm{RMS} = 22.98$ against $\mathrm{SD} = 17.10$, so $\rho = 1.34$.

Reading the statistic. $\rho \approx 0$ is a straight line at any slope; 0.3–0.7 is a smooth curve, a trend with gentle and consistent curvature; $\rho \approx 1$ means a typical bend is as large as the spread of the whole series; and $\rho > 1.3$ means the trajectory reverses direction at a scale rivalling the signal itself. Across the 82 words the statistic ranges from 0.16 (*woke*) to 2.72 (*they ate*), with a median of 1.23 (Table 3).

Two properties bear on any attempt to use ρ as a selection criterion. First, it is **blind to sampling noise**: a rare word is jagged because its per-year counts are small, not because usage moved, and the statistic cannot tell those apart. In practice ρ therefore tracks frequency—Spearman $\rho_s = -0.45$ against log total hits, with median roughness falling from 1.46 in the least-frequent third of the set to 0.78 in the most-frequent third—and the roughest trajectories belong to the sparsest words (*they ate*, 201 hits; *sus*, 138; *npc*, 136). A cutoff on ρ alone would select for corpus frequency about as much as for shape, and should be paired with a noise-floor comparison (per-year relative standard error $1/\sqrt{k}$) before it is read as evidence about language. Second, with $n = 12$ there are only ten second differences, so a single anomalous year can dominate: the 2024 slice is truncated at dump `2024-10` (103.9 M tokens against 1,429.7 M in 2017, Table 4), and for *aura* the 2024 term alone contributes 52% of $\sum(\Delta^2 x)^2$.

C Experiment 1: FineWeb Corpus Frequency

C.1 Corpus and token budget

The corpus ground truth is the public FineWeb (Penedo et al., 2024) `sample-10BT` config on HuggingFace: 95 dated Common Crawl (`CC-MAIN`) dumps spanning `2013-20` through `2024-10`, totalling 1.037×10^{10} tokens. Each document's `dump` field is mapped to the Monday of its ISO year-week to place it on a calendar axis, and the per-dump token totals (cached in `fineweb_10BT_dump_sizes.json`) serve as the denominators for rate normalisation. The sample is unevenly distributed across years—the mid-2010s crawls are the largest and the truncated 2024 slice the smallest—so per-year token totals (Table 4) are essential for interpreting raw counts.

C.2 Context extraction

For each target word we stream the sample and, for every surviving document, extract a ± 20-token window of the original text around each match (`fineweb_context.py`, `-context-window 20`), preserving the raw casing, punctuation, and spacing needed for sense classification. Matching is performed on normalised tokens (whitespace-split, lower-cased, leading/trailing punctuation stripped) via a regex that mirrors the tokeniser's word boundaries, so multi-word targets (*red pill*, *glow-up*) and inflected forms are captured. One CSV of `(target, uri, target_context)` rows is produced per dump.

C.3 Sense disambiguation

Each extracted context is scored with a RoBERTa classifier fine-tuned to separate the *slang* sense from the *standard-English* sense (e.g. *sick* ill vs. excellent, *aura* energy-field vs. Gen-Z coolness). The judged word is injected into a prompt so the decision is conditioned on *which* word is in question: `Is the word "{target}" used as slang here? {context}`. The paper's figures count only occurrences scoring ≥ 0.99; the threshold is a tunable parameter exposed by `roberta_filter.py` (raw scores lie in $[-1, 1]$). Appendix D documents the classifier's data and evaluation.

C.4 Normalisation and uncertainty

Sense-filtered counts are converted to occurrences per million tokens of the corresponding dump, $r_{w,d} = 10^6 \cdot \text{count}_{\text{slang}}(w, d)/\text{tokens}(d)$, using the per-dump token totals above. Each per-dump count is modelled as a Poisson draw $k \sim \text{Poisson}(r_{w,d} \cdot \text{tokens}(d))$, and the plotter can shade the exact (Garwood) Poisson interval (95% by default) as a band that tightens as the centred smoothing window aggregates more of the sample. The band figures in this paper are drawn without that shading (`-no-ci`), which at 95 dumps and up to eight overlapping words obscured the lines it annotated.

C.5 Plotting and peak extraction

The band figures (Figure 1) use a linear y-axis rescaled per word to a percentage of that word's peak smoothed dump (the `_pct` variant, `-percent-of-peak`) with a 9-dump centred moving average and no shaded CI band. The rescaling removes the need for the broken y-axis the earlier `_count` figures used to stop *lol* compressing the other curves. Dumps with zero sense-filtered occurrences are dropped so the line connects neighbouring dumps rather than falling to the floor, and a $\log_2$-axis version of each figure is also produced. The corpus peak year (Table 1) is the argmax, over calendar years, of the year-pooled normalised rate (`peak_year.py`). All band figures are generated by `word_rate_plotter.py -threshold 0.99 -highlight-bands -percent-of-peak -no-ci -clean -settings-note -smooth 9`, which pools the target and scenario scored contexts.

C.6 Implementation

`FineWebAnalysis/`: `fineweb_context.py` (context extraction), `roberta_filter.py` (context scoring), `word_rate_plotter.py` (frequency curves and Poisson bands), `peak_year.py` (peak-year extraction).

D Sense Classifier: Data, Training, and Validation

D.1 Model and training

The sense classifier is a `roberta-base` sequence classifier fine-tuned for binary slang-vs-standard judgement (`finetune_roberta.py`). The target word is injected into the prompt template above and the rendered string is truncated/padded to 256 tokens. Training uses a class-weighted cross-entropy loss (weights inversely proportional to class frequency, to offset the label imbalance), AdamW (learning rate 2×10^{-5}, weight decay 0.01), batch size 16, and a linear schedule; we hold out a stratified 20% split (`random_state=42`) and keep the

checkpoint with the best validation accuracy. The exact prompt template is serialised next to the model so that `roberta_filter.py` applies identical wording at inference.

D.2 Annotation data

Training data combines human-annotated FineWeb contexts (1,310 rows over 15 words, 649 slang / 661 standard) with a larger synthetically generated set (8,504 rows over 124 words (104 unique slang + 21 common forms, e.g. yap, yapping or rage bait, rage-bait), 6,108 slang / 2,396 standard). Synthetic examples are produced by prompting Claude Opus 4.7 for 50–100 natural example sentences per target word, each self-labelled slang or standard; words with a genuine everyday sense are balanced ∼50/50, while slang-only words (*skibidi*, *rizz*, . . .) are all-slang (`generate_synthetic_annotations.py`). Conditioning the classifier on the target word lets a single model generalise across the full target vocabulary rather than requiring a per-word classifier. This set is manually validated by sampling. Out of 8504 text rows generated, we randomly sampled 1000* and manually validated Claude's annotations. Human annotator (me) agreed in 98.69% of cases, and marked 1 (0.38%) as false positives for slang and 12 (1.64%) as false negatives for slang usage of the target word.

D.3 Validation of the synthetic labels

Because the training set leans on synthetic labels, we audited their reliability against human judgement on a 995-example sample (`validation_metrics.py`), treating the manual annotation as ground truth. Synthetic labels agree with human labels 98.69% of the time (982/995 samples). Other metrics: precision 99.86%, recall 98.36%, $F_1 = 99.11\%$; false-positive rate 0.38%, false-negative rate 1.64%; confusion matrix TN = 261, FP = 1, FN = 12, TP = 721, confirming the synthetic labels are a faithful stand-in for human sense judgements.

D.4 Held-out evaluation protocol

The trained classifier is evaluated on a separate, class-balanced held-out set of FineWeb contexts (1,340 rows over 15 words, 670 slang / 670 standard; `fine_web_small_validation_set`) via `auc_eval.py`, which reports ROC AUC together with a confusion matrix, precision/recall/F_1 at the operating threshold, and a per-word breakdown. Passing `-train-dir` additionally restricts the metrics to contexts whose text does not appear in the training annotations, isolating performance on genuinely unseen strings. On this held-out set, the final `ft_model_roberta` checkpoint attains an overall ROC AUC of **0.9906**; at the paper's operating threshold (≥ 0.99, i.e. `ft_score` ≥ 0.98) precision is **94.44%** and recall is **96.42%** ($F_1 = 95.42\%$; confusion matrix TN = 632, FP = 38, FN = 24, TP = 646). Restricting to the 1,234 rows (92.1%) whose context does not overlap the training annotations (`completed_annotations`) via `-train-dir` gives an unseen-only ROC AUC of **0.9924**, precision **94.48%**, and recall **96.77%** ($F_1 = 95.61\%$), confirming the classifier's performance is not an artefact of train/validation leakage.

E Experiment 2: Direct Year Association

These queries share the system message `"Give the response as a single returned token or word."` Two are free-text point estimates and the remainder are logprob/echo queries over the same nineteen target words and the year range 2013–2025:

1. **GiveYear:** "The slang word '{word}' is most associated with what year? Respond using only the year."
2. **DatePhrase:** "You hear someone saying '{word}' in conversation, which year are they most likely saying this in?"
3. **GiveTopSlang:** "What is the most unique and popular slang word associated with the year {year}? Respond using only a single word." — the inverse mapping (year → word), collected as free text asnd parsed; reported as supplementary.
4. **EraAppropriate:** "It's {year}, is it appropriate to use the term '{word}'?" — a year-conditioned appropriateness judgement swept over 2013–2025 for each target word. Unlike GiveYear/DatePhrase, which ask the model to *name* a year, this experiment fixes the year in the prompt and asks the model to *judge* whether the term fits it, testing whether the model treats a slang word as dated once it is past its era.

GiveYear and DatePhrase are rendered once per target word and sampled five times per word on all

four models ($19 \times 5 \times 2 \times 4 = 760$ responses); a four-digit year is parsed from each response. The mean responded year per word, pooled over GiveYear and DatePhrase, is plotted against the corpus peak in Figure 2; the per-model mean absolute errors against the corpus peak are 6.5 (Claude), 5.5 (GPT-4o, inflated by its collapse to "2023"), 10.3 (DeepSeek), and 7.1 (Llama) years.

E.1 EraAppropriate: year-conditioned appropriateness judgement

The EraAppropriate prompt is swept over all nineteen target words and the thirteen prompt years 2013–2025, sampled five times per (word, year) pair on all four models ($19 \times 13 \times 5 \times 4 = 4{,}940$ responses). Each free-text answer is classified as *yes* (appropriate), *no* (inappropriate/dated), or *unclear* by a lexical parser keyed on the leading yes/no token; the unclear rate is negligible ($\leq 0.4\%$ for every model). For each model we plot the fraction of decided responses answering "yes" as a heatmap of target word (y-axis, ordered by true corpus peak) against prompt year (x-axis), with each word's true corpus peak outlined (Figure 8). A model that tracked usage would answer "yes" up to roughly a word's peak year and switch to "no" afterward.

Instead, the models judge slang to be perennially appropriate: the overall yes-rate is 100.0% for `gpt-4o`, 96.7% for `claude-sonnet-4-6`, 88.4% for `DeepSeek-V4-Pro`, and 78.6% for `Llama-3.3-70B`. Critically, appropriateness does not fall off after a word's true peak. Splitting each word's cells into pre/at-peak versus post-peak years, the yes-rate is essentially unchanged for Claude ($96.4\% \rightarrow 97.2\%$) and GPT-4o ($100\% \rightarrow 100\%$) — not one of the nineteen words shows a post-peak decline — and it actually *rises* for DeepSeek ($85.9\% \rightarrow 91.8\%$). Only `Llama-3.3-70B` shows a modest dating effect ($84.0\% \rightarrow 71.6\%$, with 7 of 19 words declining after their peak), consistent with its generally lower and noisier yes-rate. This is a third line of evidence, alongside the backward-shifted year associations above and the year-cue-insensitive recovery of Experiment 3, that the models lack an actionable sense of when a slang term is era-appropriate: they will neither stop reaching for dated slang nor, when asked directly, flag it as out of date.

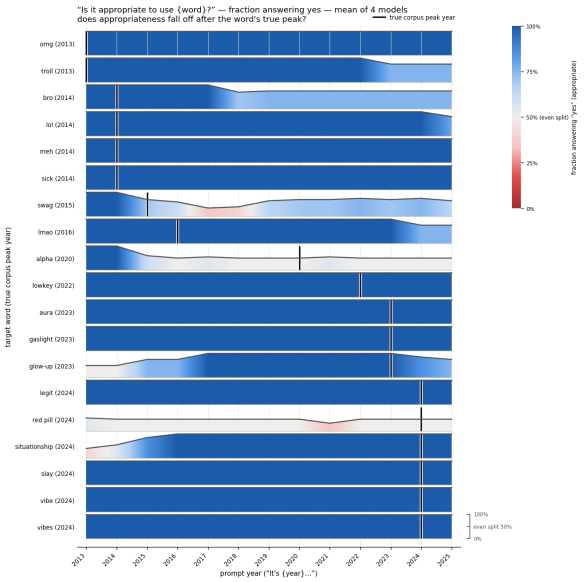


Figure 8: EraAppropriate judgements for `claude-sonnet-4-6`: each cell is the fraction of responses answering "yes, appropriate" to "It's {year}, is it appropriate to use the term '{word}'?", for target word (y-axis, ordered by true corpus peak year, shown in parentheses) against prompt year (x-axis; five samples per cell). Green is "yes", red is "no"; each word's true corpus peak year is outlined. A model tracking usage would show a green band up to the peak turning red afterward; instead the panel is almost uniformly green, i.e. the model rarely judges any slang term inappropriate for any year and does not treat terms as dated past their peak. Per-model panels (`era_appropriate_*.png`) show the same near-uniform pattern, with `Llama-3.3-70B` the only model exhibiting a partial post-peak drop-off.

F Experiment 3: Semantic Slot Querying

All slot prompts are sampled at temperature 1.0. Concept-slot prompts use the system message `"Respond with a single word only. No explanation, no punctuation - just the word."`; target-recovery prompts use `"Respond with only a single word or very short phrase - no explanation, no punctuation."`

Concept-slot mode. Six meaning slots, each with four independently worded phrasings (24 prompts total); the model's single-word answer is recorded without naming any target, surfacing its default vocabulary per slot. The full phrasing set:

1. *cool/impressive:* (1) "What's a single informal word someone might use to describe something they think is very impressive or cool?" (2) "Give me one slang word people

use to say something is excellent or impressive." (3) "What's an informal word for something that is really good and worth getting excited about?" (4) "Name a single casual word that means something is exceptionally cool or impressive."

2. *mediocre:* (1) "What's a single informal word for something that is average, unimpressive, or not worth your time?" (2) "Give me one slang word people use to say something is mediocre or disappointing." (3) "What's a casual word that means something failed to meet expectations?" (4) "Name a single informal word that means something is just okay — not good, not terrible."

3. *charisma:* (1) "What's a single informal word for someone's natural ability to effortlessly attract others or come across as charming?" (2) "Give me one slang word that describes the quality of being naturally magnetic, charming, or romantically appealing." (3) "What's a casual word for the thing that makes someone just naturally attractive or charismatic without trying?" (4) "Name a single informal word for the kind of confident, effortless charm that makes people drawn to someone."

4. *agreement:* (1) "What's a single informal word someone uses to express strong enthusiastic agreement or approval?" (2) "Give me one slang word people say to mean 'yes, absolutely, that's totally true.' " (3) "What's a casual word for when you hear something and you completely agree with it?" (4) "Name a single informal word used to affirm what someone just said — like a stronger, more casual 'exactly.' "

5. *suspicious:* (1) "What's a single informal word for when something about a person or situation feels off, untrustworthy, or not right?" (2) "Give me one slang word that means something seems off or like it shouldn't be trusted." (3) "What's a casual word for when you sense that someone is being deceptive or hiding something?" (4) "Name a single informal word for a feeling that something is unsafe or that a person can't be trusted."

6. *excellent (sensory):* (1) "What's a single informal word someone would use to say a song, food, or piece of art is exceptionally good?" (2) "Give me one slang word people use when something sounds, tastes, or feels really, really good." (3) "What's a casual word for when a song or meal is so good it gives you a visceral, satisfying reaction?" (4) "Name a single informal word meaning something is so good it almost overwhelms you in a great way."

Target-recovery mode. Each of the nineteen target words has three phrasings written to evoke that word specifically. A response counts as a recovery if it matches the target after case-folding, applying a light suffix stemmer and whole-word phrase containment — so *glowup*/*glow-up*, *gaslighting*/*gaslight*, and "take the red pill"/*red pill* all count, while *swagger* is not credited as *swag*. Listing 1 shows two of the nineteen templates.

Year-conditioned variant. Each target-recovery prompt is prefixed with "The year is {year}." and the year is swept over 2013–2025 (13 values). With four models, nineteen words, three phrasings, thirteen years, and ten samples per cell this yields 29,640 responses (7,410 per model; 30 per word × year cell) — the data behind the year-resolved heatmaps (Figure 4). Aggregate target-word recovery (Figure 3) pools these responses over the stated year.

Listing 1: Two of the nineteen target-recovery prompt templates (Experiment 3). The year-conditioned variant sweeps the `year` list over 2013–2025; the unconditioned recovery figure pools over all years.

```
{ "id": "exp3_lol",
  "system": "Respond with only a single word or very short phrase -- no explanation, no punctuation.",
  "user": "The year is {year}. {phrasing}",
  "temperature": 1.0,
  "variables": {
    "year": ["2013", "...", "2025"],
    "phrasing": [
      "What's a way, using text, people might show that something made them laugh?",
      "Give me the short response someone adds to a message to indicate they're laughing.",
      "What phrase do people use online to react to something funny?"
    ]
  }
}

{ "id": "exp3_aura",
  "system": "Respond with only a single word or very short phrase -- no explanation, no punctuation.",
```

```
  "user": "The year is {year}. {phrasing}",
  "temperature": 1.0,
  "variables": {
    "year": ["2013", "...", "2025"],
    "phrasing": [
      "What's a single informal word for the intangible
       coolness or presence someone radiates?",
      "Give me one word young people use for the invisible
       'points' you gain or lose by acting cool or cringe.",
      "What word for an invisible energy field around a
       person is now slang for their coolness or charisma?"
    ]
  }
}
```

G Models and Sampling

1. `meta-llama/Llama-3.3-70B-Instruct-Turbo` — Together AI
2. `deepseek-ai/DeepSeek-V4-Pro` — Together AI
3. `gpt-4o` — OpenAI
4. `claude-sonnet-4-6` — Anthropic

A single harness (`PromptingSlang/`) dispatches each prompt to the correct provider by model-id convention (`org/model` → Together, `claude-*` → Anthropic, `gpt-*/o#-*` → OpenAI), retrying with exponential backoff. Prompts expand on their `variables` block, and each expansion is sampled `-samples` times; generation is capped at 512 tokens. All querying experiment prompts set temperature 1.0 per template. Because the closed APIs do not expose token logprobs, their response distributions are approximated by repeated sampling. Direct logprob/echo queries were intially run on the open models, but these were dropped in favor or consistency between the two model types. Runs are keyed by `(model, prompt, sample index)` and are resumable, so partial runs top-fill rather than restart. Prompt templates and the response-collection harness are in the `PromptingSlang/` directory of the project repository.

H Naturalistic Generation: Statistical Analyses

This appendix details the persona-simulation queries of Experiment 4 (Section 4.4, Figure 5) and the estimators that quantify its "temporal vintage."

H.1 Prompt and vocabulary

Each model is asked to role-play informal peer conversation via two prompts (system: `"You are a young person using informal language like you would among peers. . . "`; e.g. "Simulate a brief conversation between two or more teenaged friends taking place online or over text message"), yielding 100 responses per model (400 total). Generated text is matched against a 150-word slang vocabulary (the union of `target_words.txt` and `scenario_words.txt`), each word carrying the corpus peak year of Experiment 1. As in the forced-choice queries, output concentrates on a handful of high-frequency, older terms: pooled across models the top items are the skull emoji (`U+1F480`), *u*, *lol*, *omg*, and *bro*, while post-2020 slang appears with counts near 1.

H.2 Effective vintage

We summarise each model's produced slang by its *effective vintage*: the count-weighted mean corpus-peak year of the slang it emits (`effective_vintage.py`). Over the vocabulary of 88 words with ≥ 100 corpus hits (31 of which appear across the 400 pooled responses, 1,848 in-lexicon hits), the model vintage is **2014.0** (95% CI $[2013.9, 2014.1]$; parametric Poisson bootstrap, $B = 5000$, seed 42) against a frequency-weighted corpus vintage of **2015.8** — a lag of -1.7 years, i.e. models skew older than the corpus baseline over the same vocabulary. The high-frequency skull emoji (`U+1F480`) is excluded from the vintage estimators, as its corpus rate is unreliable under FineWeb's text-processing.

Word	Hits	SD	RMS	ρ	Word	Hits	SD	RMS	ρ
they ate	201	19.7	53.6	2.72	*goon*	481	28.1	31.7	1.13
sus	138	23.2	60.0	2.59	*shitpost*	113	34.1	38.4	1.13
he ate	311	19.0	48.5	2.54	*lmao*	3,816	27.9	30.9	1.11
catfish	781	21.7	54.1	2.49	*stan*	1,660	18.5	20.2	1.09
thot	442	19.5	44.9	2.31	*hella*	2,628	17.8	18.6	1.04
npc	136	29.2	66.8	2.29	*pog*	81	25.5	26.4	1.04
sigma	242	24.2	54.6	2.25	*fresh*	2,797	16.9	17.2	1.02
era	933	16.2	34.2	2.11	*swag*	2,624	20.9	20.9	1.00
unc	147	30.4	63.6	2.10	*cucked*	83	33.6	32.9	0.98
on god	231	17.3	36.1	2.08	*yapping*	407	28.3	27.6	0.98
extra	680	14.3	28.4	1.99	*vibing*	564	26.0	25.2	0.97
mog	93	29.4	57.9	1.97	*rad*	5,490	17.2	16.1	0.93
she ate	198	16.5	31.5	1.91	*ick*	559	28.9	25.8	0.89
bruh	635	19.8	36.4	1.84	*goals*	432	30.9	27.5	0.89
cooked	1,848	15.3	27.3	1.78	*mid*	2,166	16.3	13.2	0.81
ratchet	754	23.9	42.1	1.76	*karen*	2,719	20.5	16.0	0.78
doggo	208	28.8	48.7	1.69	*meh*	5,775	23.6	18.4	0.78
brat	676	21.6	36.2	1.68	*glow-up*	98	35.7	27.6	0.77
doomscrolling	89	33.7	55.8	1.66	*6-7*	244	27.1	20.4	0.75
simp	108	31.1	49.8	1.60	*omg*	24,593	23.8	17.5	0.73
struggle bus	107	38.3	59.3	1.55	*troll*	6,077	20.4	14.5	0.71
idgaf	158	21.3	33.0	1.55	*gaslight*	316	30.3	21.4	0.71
unboxing	802	24.5	37.5	1.53	*stfu*	827	27.8	19.4	0.70
red pill	546	28.3	43.1	1.52	*bro*	7,914	21.4	14.7	0.69
dank	576	23.3	35.2	1.51	*twinning*	193	35.5	23.8	0.67
thirsty	1,500	14.4	21.6	1.50	*sick*	14,015	23.4	15.6	0.67
glazing	149	34.0	50.6	1.49	*lowkey*	567	34.9	22.8	0.65
cancelled	194	21.3	31.1	1.46	*lol*	107,305	29.6	19.2	0.65
finna	163	19.2	27.6	1.44	*tight*	359	25.7	16.1	0.63
slop	371	17.6	23.9	1.36	*situationship*	121	30.8	16.9	0.55
unbox	176	29.5	39.6	1.34	*bae*	1,649	29.4	16.1	0.55
aura	1,231	17.1	23.0	1.34	*u*	92,332	31.2	16.2	0.52
yas	361	22.7	30.2	1.33	*pumped*	8,967	18.2	8.8	0.48
slaps	1,064	21.6	28.1	1.30	*slay*	1,795	24.7	11.5	0.47
hooman	237	28.9	37.6	1.30	*methinks*	860	28.3	12.6	0.44
cuck	326	32.6	42.3	1.30	*dude*	61,357	25.0	9.8	0.39
shitposting	198	36.2	46.5	1.29	*yeet*	99	30.8	11.6	0.38
`U+1F480`	149	32.5	41.5	1.28	*lmfao*	1,283	28.4	9.9	0.35
catfished	164	28.2	36.0	1.28	*legit*	32,897	22.7	7.6	0.33
griefing	165	24.4	30.2	1.23	*vibes*	20,311	25.7	7.6	0.29
yap	387	25.6	31.5	1.23	*woke*	1,555	35.5	5.5	0.16

Table 3: Trajectory roughness for the 82 words passing the ephemerality screen ($\geq$ 80 slang-sense hits and a minimum year at or below 50% of peak), ordered by ρ descending and continuing in the right-hand block. "Hits" is the total count of occurrences scoring ≥ 0.99 on the slang-sense classifier; SD is the sample standard deviation of the percent-of-peak series; RMS is the root-mean-square of its ten second differences; ρ is their ratio (Eq. 1). Values are computed by `FineWebAnalysis/trajectory_explorer.py` from `peak_years_pct.json`. The skull-emoji target is written `U+1F480`.

Year	Tokens (M)	Year	Tokens (M)
2013	185.4	2019	1,236.9
2014	714.1	2020	974.6
2015	827.7	2021	1,134.4
2016	765.7	2022	852.1
2017	1,429.7	2023	809.9
2018	1,336.9	2024	103.9

Table 4: FineWeb 10BT sample token totals by calendar year (millions), summed over that year's `CC-MAIN` dumps. Total: 10,371.5 M tokens across 95 dumps. The 2024 slice is truncated at dump `2024-10`.